for communication. Our next step is to modify the above code and create the next version that detects the language automatically. For this, we need another module called "langdetect." To install the module, run the following command in the terminal:

```
sudo pip3 install langdetect
```

EFY Note
The source code of this project is available for free download at *www.electronicsforu.com*

```
File  Edit  Format  Run  Options  Window  Help
import speech_recognition as sr
import math
import time
import sys
import openai
import pygame
from gtts import gTTS
pygame.mixer.init()
import pyttsx3
from langdetect import detect

#model_to_use="text-davinci-003" # most capable
#model_to_use="text-curie-001"
#model_to_use="text-babbage-001"
model_to_use="text-davinci-003" # lowest token cost
language_code= "hi"
r = sr.Recognizer()
#openai.api_key="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"
#openai.api_key="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"
openai.api_key="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"

def chatGPT(query):
```

Fig. 5: Importing language detection module

```
def chatGPT(query):
        response = openai.Completion.create(
                model=model_to_use,
                prompt=query,
                temperature=0.7,
                max_tokens=100,
                n=1,
                stop=None,
                )
        return str.strip(response['choices'][0]['text']), response['usage']['tot
def detect_language(text):
    return detect(text)
```

Fig. 6: Creating language detection function

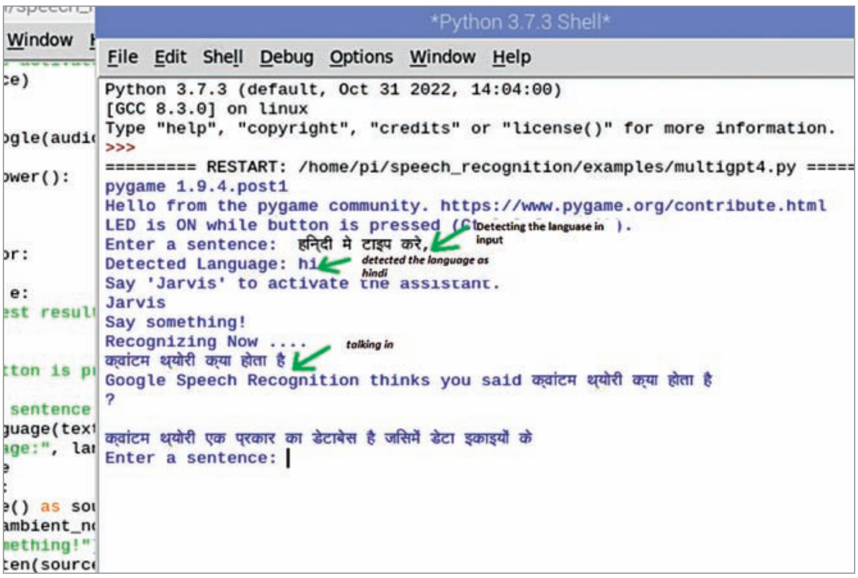

Fig. 7: Code used for testing

Now, in the code, we import the "langdetect" Python module along with OpenAI and another Python module. We then create another function. Fig. 5 illustrates the process of importing the language detection module.

Following this, we create another function named "detect_language" that prompts for text input and automatically detects the language in the input query. You can input text in any language, such as Korean, Devanagari, Marlin, Latin, etc, and the function will automatically detect the language. Fig. 6 demonstrates the creation of the language detection function.

After language detection, we set the language code to the detected language, using the value determined by the "detect_language" function. Within the while loop, we capture audio, extract the human voice, and use NLP in the detected language to convert it into text.

This text is then sent to ChatGPT as a query, and NLP converts the response into voice in the same language. Congratulations, your code is now ready, and the VoiceGPT speaker with multilingual support is prepared.

Testing ChatGPT voice control system

You can now run the code. It will prompt you to input a query in any language, and it will automatically detect and communicate with ChatGPT in that language. So, now you can talk to ChatGPT in any language; it will understand and reply using the same interface. Fig. 7 shows a testing code used to test this project. EFY

Ashwini Kumar Sinha is a tech journalist at EFY